

Department of Electronics
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Laboratory of Electronic Elements

Exercise 02

Diodes

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Motivation: Why does a blue LED require a higher voltage than a red one? Why does the Zener diode stabilize voltage so effectively? By the end of this lab you will be able to answer these questions using real measurement data.

1. Goal of the Exercise

- Measure DC current-voltage (I-V) characteristics of rectifier, LED, and Zener diodes.
- Get familiar with practical aspects of diode operation.
- Determine and estimate the basic parameters of semiconductor junctions.

2. Equipment and Components

Equipment

- NI ELVIS Prototyping Board connected to a PC
- Virtual Instruments (VI):
 - Digital Multimeter (DMM)
 - Variable Power Supply (VPS)
 - Two-Wire Voltage Analyzer (2-Wire)

Components

Type	Values / Part numbers
Resistors	100 Ω , 220 Ω , 330 Ω , 470 Ω , 1 k Ω
Rectifier diodes	1N4001, 1N4448, 1N914, BAV21
Zener diodes	3.3 V, 5.1 V, 9.1 V
LEDs	Red, Green, White, Blue, Yellow

3. Theoretical Background

Read this section before/after the lab. Understanding the physics behind the diode will help you interpret your measurements — not just record numbers.

3.1 The p-n Junction

A **semiconductor diode** is a two-terminal device that conducts current primarily in one direction. It is based on a **p-n junction**: the interface between a p-type semiconductor (majority carriers: holes) and an n-type semiconductor (majority carriers: electrons). At the junction, a **depletion region** forms, free of mobile charge carriers.

- **Forward bias** (anode at higher potential): the barrier decreases, the depletion region narrows, and current flows easily.
- **Reverse bias** (cathode at higher potential): the barrier increases, the depletion region widens, and only a tiny leakage current flows — until breakdown.

3.2 The Shockley Equation

The ideal I-V characteristic of a diode is described by the **Shockley equation**:

$$I_D = I_S \left(e^{\frac{U_D}{nV_T}} - 1 \right) \quad (1)$$

where:

- I_D — diode current [A]
- I_S — **saturation current** (reverse leakage current), typically $\sim$ pA for silicon
- U_D — voltage across the junction [V]
- n — **ideality (nonideality) factor** of the junction ($1 \leq n \leq 2$): $n = 1$ for ideal diffusion current, $n = 2$ for recombination-dominated current
- $V_T = \frac{kT}{q} \approx 26 \text{ mV}$ at $T = 300 \text{ K}$ — **thermal voltage**

3.3 The I-V Characteristic in Linear and Lin-Log Scale

Linear scale: Three operating regions are visible:

1. **Reverse region** ($U_D < 0$): current $\approx -I_S$ (negligible).
2. **Threshold / turn-on voltage** U_{th} : below this, current is negligible. For silicon: $U_{th} \approx 0.5\text{--}0.6 \text{ V}$.
3. **Forward region** ($U_D > U_{th}$): current rises exponentially.

Lin-log scale (log I vs. U): Taking the natural logarithm of Eq. (1) (for $I_D \gg I_S$):

$$\ln(I_D) = \ln(I_S) + \frac{U_D}{nV_T} \quad (2)$$

The characteristic becomes **linear**, with:

- Slope = $\frac{1}{nV_T} \rightarrow$ determines n
- Intercept = $\ln(I_S) \rightarrow$ determines I_S
- At high currents, the slope decreases due to **series resistance** R_S (bulk + contact resistance)

3.4 Key Junction Parameters

Symbol	Name	Description
I_S	Saturation current	Reverse leakage, typically pA–nA for Si
n	Ideality factor	Describes recombination mechanism; $n=1$: diffusion, $n=2$: recombination
R_S	Series resistance	Ohmic resistance in bulk; visible as slope reduction at high currents in lin-log
I_{gen}	Generation current	Recombination current component ($\propto e^{U_D/2V_T}$)

3.5 Small-Signal (Differential) Resistance

For small AC signals superimposed on a DC bias point (U_0, I_0) , the diode behaves as a **small-signal resistance** r_d :

$$r_d = \frac{1}{g_d} = \left(\frac{dI}{dU} \right)^{-1}_{(U_0, I_0)} = \frac{nV_T}{I_0} \quad (3)$$

Graphical method: measure ΔU and ΔI at two points symmetrically around the bias point. Then $r_d = \Delta U / \Delta I$.

3.6 Zener Diodes

A **Zener diode** is designed to operate in the reverse breakdown region. Two mechanisms cause breakdown:

- **Zener effect** (quantum tunnelling): dominant for $V_Z < 5\text{ V}$; negative temperature coefficient of V_Z .

- **Avalanche breakdown:** dominant for $V_Z > 7\text{ V}$; positive temperature coefficient of V_Z .
- For $V_Z \approx 5\text{--}7\text{ V}$: mixed mechanism, nearly zero temperature coefficient $\rightarrow$ best for precision voltage references.

The **dynamic (small-signal) resistance** $r_Z = \Delta U / \Delta I$ determines the quality of voltage stabilization: **lower** r_Z = **better stabilizer**.

3.7 LEDs (Light-Emitting Diodes)

LEDs are p-n junction diodes where electron-hole recombination releases energy as **photons**. The emitted wavelength λ is determined by the bandgap energy E_g of the semiconductor:

$$\lambda = \frac{hc}{E_g} \quad (4)$$

The forward voltage scales with E_g , hence with the emitted color:

Color	λ [nm]	Typical U_F [V]
Infrared	> 760	~ 1.2
Red	620–760	1.6–2.0
Yellow	570–620	1.9–2.2
Green	495–570	2.0–3.0
Blue	450–495	2.5–3.5
White	broadband	2.8–3.5

4. Pre-Lab Draft — *Can you complete all these tasks before the lab?*

Task P1. Draw an **ideal** and a **real** DC I-V characteristic of a rectifier diode. Show the real characteristic in:

- linear scale (both axes linear)
- lin-log scale (x-axis: voltage; y-axis: $\log I$)

Hint: Recall the Shockley equation (Eq. 1).

Task P2. On your lin-log I-V plot, identify and label the following regions/parameters:

- Region dominated by recombination current ($n \approx 2$)
- Region dominated by diffusion current ($n \approx 1$)
- High-current region affected by series resistance R_S
- Parameters: saturation current I_S , ideality factor n , series resistance R_S

Describe each parameter briefly (1–2 sentences).

Task P3. Using datasheets (links in Section 7), draw the reverse-bias I-V characteristics of the three Zener diodes: $V_Z = 3.3\text{ V}$, $V_Z = 5.1\text{ V}$, $V_Z = 9.1\text{ V}$. Plot all three on one graph. Label the breakdown voltage for each.

Task P4. Based on the table in Section 3.7, predict the forward voltage U_F for each LED color used in this exercise (Red, Green, White, Blue, Yellow). Fill in the table below.

Color	λ [nm] (predicted)	U_F [V] (predicted)	U_F [V] (measured)
Red			
Yellow			
Green			
Blue			
White			

Task P5. Define the small-signal resistance r_d of a diode. Describe (in writing, 3–5 sentences) both the analytical and graphical method for its determination. Write down the formula: $r_d = \dots$

5. Course of the Exercise

Hint

Equipment setup reminder:

NI ELVIS instruments:

- **VPS (Variable Power Supply):** voltage is set via the software knob in the ELVIS interface. The VPS output terminals (supply voltage and ground) are available directly on the **Prototyping Board**. Note: VPS does *not* have a current compliance setting — current limiting must be ensured by the external circuit (series resistor).
- **2-Wire Analyzer:** connect the component between DUT+ and DUT− terminals on the Prototyping Board. Set voltage range, step size, and current limit before running.

Other equipment:

- **DMM (Agilent 34401A):** use the bench multimeters available at the station — **not** the ELVIS built-in DMM. Connect using **banana cables**; use the **banana-to-prototyping-board adapters** mounted on the ELVIS unit to access the circuit on the Prototyping Board. For voltage measurement, connect in parallel with the component; for current measurement, connect in series with the circuit. Set to **DC V** or **DC I** mode as required.
- Never exceed absolute maximum ratings! For LEDs: $I_{max} = 25\text{ mA}$.

Part A: Rectifier Diodes

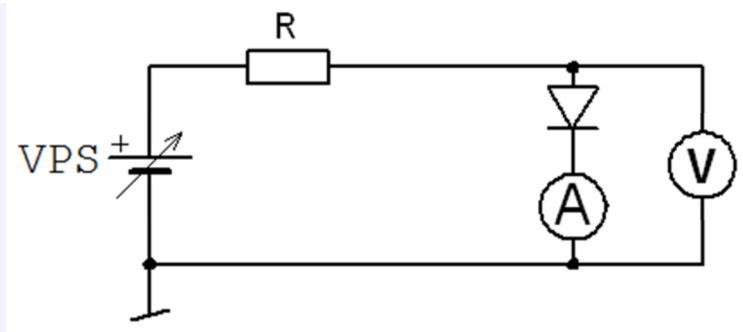
Task 1 — Point-by-point I-V measurement

Component: rectifier diode selected by the instructor (1N4001 or 1N4448).

Before you start: Find and open the datasheet for your assigned diode (links in Section 7). Note the following parameters and write them below — you will compare them with your measured values later:

Parameter	Datasheet value
Maximum forward current $I_{F,max}$ [mA]	
Forward voltage U_F at $I_F = 10\text{ mA}$ [V]	
Reverse leakage current I_R [μA]	

Circuit: Connect the diode in series with a $330\ \Omega$ resistor between the VPS output (Prototyping Board terminal) and ground. Use the two **Agilent 34401A** bench multimeters available at the station to measure voltage and current simultaneously, as described below.



Measurement circuit for Task 1: DMM1 measures voltage across the diode; DMM2 measures the series current.

DMM 1 — Voltmeter (measuring U_D):

- Press **DC V** on the front panel.
- Connect the **red** probe to the **HI** terminal (upper right, labelled *Input $V\Omega \rightarrow$*).
- Connect the **black** probe to the **LO** terminal (lower centre).
- Connect the probes **in parallel** across the diode: red to the **anode**, black to the **cathode**.
- Leave the range on **Auto**, or select manually with $\wedge/\vee$: **1 V** range is suitable for forward bias.

DMM 2 — Ammeter (measuring I_D):

- Press **Shift** then **DC V** to select the **DC I** function (the label *DC I* is printed in blue above the *DC V* button).
- Connect the **red** probe to the **I** terminal (lower right, labelled *3 A / Fused on Rear Panel*).
- Connect the **black** probe to the **LO** terminal (lower left or lower centre — both are internally connected).
- Insert the ammeter **in series** with the circuit by breaking the wire between the 330Ω resistor and the diode anode.
- Recommended ranges: **20 mA** for forward bias; switch to **100 μ A** before applying reverse bias. You may also leave the range on **Auto**.

Calculate the diode current from the voltage drop across the resistor as a cross-check: $I_D = U_R/330\Omega$, where $U_R = U_{VPS} - U_D$.

CAUTION

Note that VPS has no built-in current limiting. The $330\ \Omega$ series resistor keeps the current in a safe range — as long as U_{VPS} stays within 1.0 V. No need to go higher for this task.

Procedure:

1. Set VPS output to 0 V. Enable output.
2. Increase voltage step by step as follows:
 - **Forward bias:** steps of 0.1 V from 0 to 0.4 V; then steps of 0.05 V from 0.4 to 0.7 V; then steps of 0.1 V up to 1.0 V maximum.
 - **Reverse bias:** steps of 1 V from -1 to -5 V (read leakage current with DMM in μA mode).
3. At each step record: U_{VPS} , U_D (voltage across diode), I_D (current).

Measurement table — forward bias:

U_{VPS} [V]	U_D [V]	I_D [mA]	U_{VPS} [V]	U_D [V]	I_D [mA]
0.0			0.50		
0.1			0.55		
0.2			0.60		
0.3			0.65		
0.4			0.70		
0.45			0.80		
—			0.90		
—			1.00		

Measurement table — reverse bias:

U_{VPS} [V]	U_D [V]	I_D [μA]	U_{VPS} [V]	U_D [V]	I_D [μA]
-1			-4		
-2			-5		
-3			—		

Data entry: Transfer all measured values into a spreadsheet (Excel). Add a column for $\ln(I_D)$ — you will need it for the lin-log plot in Task 5.

Task 2 — Automatic I-V measurement with 2-Wire Analyzer

Components: all rectifier diodes selected by the instructor.

Procedure (for each diode):

1. Connect diode between **DUT+** (anode) and **DUT−** (cathode).
2. Open the *Two-Wire Current-Voltage Analyzer* in ELVIS.

3. Set the parameters:

- *Start Voltage*: -1.0 V
- *Stop Voltage*: $+1.0\text{ V}$
- *Step Size*: 5 mV
- *Current Limit*: 20 mA
- *Gain*: Low

4. Click **Run**. Export / screenshot the characteristic.

5. Repeat for all diodes assigned by the instructor.

Record measured diodes: _____

Part B: Zener Diodes

Task 3 — Zener diode I-V characteristics

Components: Zener diodes $V_Z = 3.3\text{ V}$, 5.1 V , 9.1 V .

CAUTION

Connect Zener diodes the **same way as rectifier diodes (anode to DUT+, cathode to DUT-)**. The 2-Wire Analyzer will sweep from negative to positive voltage; the Zener breakdown appears on the **left side** of the characteristic (negative U_{AK} , i.e. reverse bias). Do not exceed I_{max} from the datasheet.

Procedure (for each Zener diode):

1. Connect the Zener diode between DUT+ (**anode**) and DUT- (**cathode**).
2. Open the *Two-Wire Analyzer*. Set:
 - *Start Voltage*: $-2 \times V_Z$ of the measured Zener (i.e. -6.6 V , -10.2 V , -15 V for the 3.3 V , 5.1 V , and 9.1 V diode respectively; limited to -15 V maximum)
 - *Stop Voltage*: $+1.0\text{ V}$
 - *Step Size*: 50 mV
 - *Current Limit*: 40 mA
 - *Gain*: Low
3. Click **Run**. Export / screenshot the characteristic.
4. Using the cursors, read the Zener voltage V_Z (the flat part of the breakdown region) and record below.

Diode	V_Z measured [V]	V_Z datasheet [V]
3.3 V Zener		3.3
5.1 V Zener		5.1
9.1 V Zener		9.1

Part C: LEDs

CAUTION

Maximum LED current: 25 mA. Always use a series resistor with LEDs. Never connect an LED directly to the VPS without current limiting! VPS has **no current compliance** — the series resistor $220\ \Omega$ is the only protection.

At $U_{VPS} = 5\text{ V}$ and $U_F \approx 2\text{ V}$: $I_{max} = \frac{5 - 2}{220} \approx 14\text{ mA}$ — safe. Do **not** exceed $U_{VPS} = 5.0\text{ V}$.

Task 4 — LED forward voltage measurement

Components: LEDs selected by the instructor (Red, Green, White, Blue, Yellow).

Procedure (for each LED):

1. Connect the LED in series with a $220\ \Omega$ resistor between the VPS output (Prototyping Board) and ground.
2. Connect the Agilent 34401A (DC V mode) directly across the LED.
3. Set VPS to the first voltage from the table below. Enable output.
4. Read U_D from the DMM. Calculate the current using:

$$I_{LED} = \frac{U_{VPS} - U_D}{220\ \Omega}$$

5. Record U_D and calculated I_{LED} in the table.
6. Repeat for each U_{VPS} value. Do **not** exceed $U_{VPS} = 5.0\text{ V}$ (ensures $I_{LED} < 25\text{ mA}$).

U_{VPS} [V]	Red		Yellow		Green	
	U_D [V]	I_{LED} [mA]	U_D [V]	I_{LED} [mA]	U_D [V]	I_{LED} [mA]
2.0						
2.5						
3.0						
3.5						
4.0						
5.0						

U_{VPS} [V]	Blue		White	
	U_D [V]	I_{LED} [mA]	U_D [V]	I_{LED} [mA]
2.0				
2.5				
3.0				
3.5				
4.0				
5.0				

Report: For each LED, note the forward voltage U_D at $I_{LED} \approx 10$ mA (interpolate if necessary). Return to the table in Task P4 and fill in the “measured” column. Compare with your predictions.

6. Data Analysis

Task 5 — Rectifier diode analysis

Using the measurement data from Tasks 1 and 2 (use a spreadsheet, e.g. Excel):

- Plot all rectifier diode I-V characteristics in **linear scale** on one graph.
- For one selected diode, plot the I-V characteristic in both **linear** and **lin-log** scale on one plot (two y-axes or two panels).
- On the lin-log graph, identify and mark:
 - Region of recombination current ($n \approx 2$)
 - Region of diffusion current ($n \approx 1$)
 - Region affected by series resistance R_S
- Extract and tabulate the following parameters for the selected diode. Compare with datasheet values.

Parameter	Measured	Datasheet
Saturation current I_S [A]		
Ideality factor n		
Series resistance R_S [Ω]		
Recombination current I_{gen} [A]		

- Discuss: How can you extract n and I_S from the lin-log plot? (3–5 sentences)

Task 6 — Zener diode analysis

Using the measurement data from Task 3:

- Plot all three Zener diode I-V characteristics (reverse-bias region) in **linear scale** on one graph. Label the Zener voltage for each diode.
- For each Zener diode, determine the **small-signal resistance** $r_Z = \Delta U / \Delta I$ at four current levels: $I = 1 \text{ mA}$, $I = 10 \text{ mA}$, $I = 20 \text{ mA}$, $I = 40 \text{ mA}$. Use the graphical (cursor) method.

Diode	r_Z at 1 mA [Ω]	r_Z at 10 mA [Ω]	r_Z at 20 mA [Ω]	r_Z at 40 mA [Ω]
3.3 V				
5.1 V				
9.1 V				

- Compare measured V_Z with datasheet values.

d) Answer in writing:

- Why do the shapes of the characteristics differ for the three Zener diodes?
- Which physical effects are responsible for these differences? (Zener tunnelling vs. avalanche)
- Which diode has the best stabilizing properties? Justify using your r_Z values.

7. References

1. Course lecture notes (P. Dziurdzia).
2. Behzad Razavi, *Fundamentals of Microelectronics*, 3rd ed., Wiley, 2021.
3. Marciniak W., *Przyrządy półprzewodnikowe i układy scalone*, WNT, Warszawa 1987.
4. 1N4001 datasheet: <https://www.vishay.com/docs/88503/1n4001.pdf>
5. 1N4448 / 1N914 datasheet (electrically equivalent to 1N4148): <https://www.vishay.com/docs/81857/1n4148.pdf>
6. BAV21 datasheet: <https://www.vishay.com/docs/85656/bav21.pdf>
7. BZX55 series (Zener diodes) datasheet: <https://www.vishay.com/docs/85604/bzx55.pdf>